

JULIE CAILLER

📍 Nancy, France | **in** /in/juliecailler | ✉ julie.cailler@loria.fr
🌐 jcailler | 🆔 0000-0002-6665-8089 | 🌐 https://jcailler.github.io/

RESEARCH EXPERIENCES

- Q Associate Professor** **Since Sept. 2024**
Loria, University of Lorraine | Nancy, France
- Member of the VeriDis research team.
 - Teaching at the Faculty of Science and Technology.
- Q Postdoctoral Researcher** **Sept. 2023 – Aug. 2024**
University of Regensburg | Regensburg, Germany
- Member of the Chair of Theoretical Computer Science.
 - Teaching assistant at the Faculty of Informatics and Data Science.
- Q Ph.D. Candidate** **Sept. 2020 – Aug. 2023**
LIRMM, University of Montpellier | Montpellier, France
- Member of the MAREL research team.
 - Teaching assistant at the Faculty of Sciences.
- Q Internship – Reliability Assessment in a Decision Support Tool** **Jun. 2019 – Aug. 2019**
INRAE | Montpellier, France
- Developpement of metrics to take in account uncertainties in user feedback.
 - Visualisation and integration of these metrics in the DOCAMEX project.
 - Survey among users to take into account their feedback and improve the tool.
- Q Intership – An Application for Multi-modal Travel** **Jun. 2018 – Jul. 2018**
LIRMM and Faciligo | Montpellier, France
- Conception and implementation of a module which matches the shortest path in multi-modal travel mode.
 - Taking into account constraints regarding the client's disabilities in the context of cotravel.

SKILLS

- ⚙ Topics** Logic, automated & interactive theorem proving, parallel programming, machine learning, quantum computing.
- ⚙ Programming Skills** Go, Scala, Python, Rocq, Ocaml, C/C++, Java, SQL, $\text{\LaTeX}$, Git
- ⚙ Languages**
- French (Mother tongue)
 - English (Professional proficiency)
 - German (Beginner)
 - Spanish (Beginner)

EDUCATION



PhD in Computer Science

2023

LIRMM, University of Montpellier | France

- Thesis topic: "Designing an Automated Concurrent Tableau-Based Theorem Prover for First-Order Logic".
- Advisors: David DELAHAYE, Hinde Lilia BOUZIANE and Simon ROBILLARD.
- Jury: Gilles DOWEK, Philipp RÜMMER, Serenella CERRITO, Damien DOLIGÉZ, Marie-Laure MUGNIER, Olivier HERMANT.
- This thesis focuses on the use of the method of analytic tableaux in the field of automatic deduction in first-order logic. In particular, it demonstrates how the use of concurrency can overcome most of the fairness challenges, improve the management of theories in tableaux and the interactions with proof assistants. These results led to the creation of the automated theorem prover Goéland.



Master Degree in Computer Science

2020

University of Montpellier | France

- Courses mainly focused on big data, artificial intelligence and natural language processing.
- Class representative.



Bachelor in Computer Science

2018

University of Montpellier | France

- Introduction courses on a wide range of Computer Science subjects, including programming, logic, graph algorithms and network.
- Class representative.



Conference
Papers

Towards Term-Based Verification of Diagrammatic Equivalence
Cailler, Julie, No   Delorme, Simon Perdrix, and Sophie Tourret
In: *13th International Joint Conference on Automated Reasoning (IJCAR 2026)*.

A Graph Rewriting-Based Semantics and Implementation for $\rho\pi$
Cailler, Julie and Martin Vassor
In: *18th Conference on Reversible Computation (RC 2026)*.

TableauxRocq: A Deep Embedding of Free-Variable Tableaux in Rocq
Rosain, Johann and Julie Cailler
In: *17th International Conference on Interactive Theorem Proving (ITP 2026)*.

Guiding Word Equation Solving Using Graph Neural Networks
Abdulla, Parosh Aziz, Mohamed Faouzi Atig, Julie Cailler, Chencheng Liang, and Philipp R  mmer
In: *Automated Technology for Verification and Analysis (ATVA 2025)*.

When GNNs Met a Word Equations Solver: Learning to Rank Equations
Abdulla, Parosh Aziz, Mohamed Faouzi Atig, Julie Cailler, Chencheng Liang, and Philipp R  mmer
In: *Frontiers of Combining Systems (FRODOS 2025)*.

Interoperability of Proof Systems with SC-TPTP
Guilloud, Simon, Julie Cailler, Sankalp Gambhir, Auguste Poiroux, Yann Herklotz, Thomas Bourgeat, and Viktor Kun  ak
In: *30th International Conference on Automated Deduction (CADE 30)*.

A Generic Deskolemization Strategy
Rosain, Johann, Richard Bonichon, Julie Cailler, and Olivier Hermant
In: *Proceedings of 25th Conference on Logic for Programming, Artificial Intelligence and Reasoning (LPAR 2024)*.

Go  land: a Concurrent Tableau-Based Theorem Prover (System Description)
Cailler, Julie, Johann Rosain, David Delahaye, Simon Robillard, and Hinde Lilia Bouziane
In: *11th International Joint Conference on Automated Reasoning (IJCAR 2022)*.



Workshop
Papers

SC-TPTP: An Extension of the TPTP Derivation Format for Sequent-Based Calculus
Cailler, Julie and Simon Guilloud
In: *9th Workshop on Practical Aspects of Automated Reasoning (PAAR 2024)*.



Posters

- Who Killed Agatha? 2022
PhD seminar | LIRMM, University of Montpellier, France
- A Concurrent Tableaux Proof-Search 2021
PhD seminar | LIRMM, University of Montpellier, France



Software

- **Goéland**
Julie CAILLER, David DELAHAYE, Isaac LLUÍS and Johann ROSAIN

Goéland is an automated theorem prover using a concurrent procedure for the tableau method for first-order logic. It is implemented in the Go programming language (with about 40 000 lines of code). As the main developer of the tool, I also supervised the different people that work or have worked on it. Goéland is available at: <https://github.com/GoelandProver/Goeland>

2022
- **TableauxRocq and POULET**
Julie CAILLER and Johann ROSAIN

TableauxRocq is a Rocq library that provides a deep embedding of free-variable tableaux, together with a fully certified fast proof checker. Both tools are available at: <https://github.com/jrosain/TableauxRocq>

2026
- **SC-TPTP Utilities**
Julie CAILLER and Simon GUILLOUD

SC-TPTP Utilities is a library of tools able to deal with the SC-TPTP format. It includes software able to handle, import, export and transform proofs in SC-TPTP format, to add intermediate proof steps, and to export them into Rocq. SC-TPTP, as well as more document about the format, is available at: <https://github.com/SC-TPTP/sc-tptp>

2022
- **G $\rho\pi$ F**
Julie CAILLER and Martin VASSOR

G $\rho\pi$ F is a web-based interactive tool for visualizing and manipulating $\rho\pi$ -calculus terms using a graph rewriting semantics. It allows you to define processes, channels, and communications, apply reduction rules, and observe the evolution of processes graphically. G $\rho\pi$ F is available at: <https://github.com/jcailler/grhopif>

2026
- **LastButNotLeast**
Julie CAILLER

LastButNotLeast is a minimalist automated theorem prover designed as a consistent baseline optimized for rapid execution and minimal resource consumption for the CASC Competition (<https://tptp.org/CASC/>), ensuring reliable performance benchmarking across competition iterations.

2025
- **CAPEX**
Julie CAILLER and Florent TORNIL

CAPEX is a knowledge-based multicriteria decision support tool that structures expert know-how to support technological decision-making in biomass transformation. It was initially applied to French cheese production under AOP/IGP, where preserving and transmitting specialized know-how is critical. By organizing this collective expertise into a structured knowledge base, CAPEX helps guide practical technological choices and supports the training of technicians and operators. CAPEX is part of the DOCAMEX project and available at: https://www.docamex.fr/docamex_portail/accueil

2019

TALKS



The ProoVer Competition

- 15th TPTP Tea Party
Lisbon, Portugal

2026
- 14th TPTP Tea Party
Stuttgart, Germany

2025



Automated Deduction Techniques

- Formal Methods Day
Loria, France

2025

	SC-TPTP : Extending the TPTP Format for Sequent-based Proofs	
	• National day of Scalp research group (Formal structures for CALculus and Proofs) <i>IRCICA, University of Lille, France</i>	2025
	• National day of LVP research group (Languages and Program Verification) <i>IRIF, University Paris-Cité, France</i>	2025
	• Formal Methods Seminar <i>Loria, Nancy, France</i>	2025
	• 13 rd TPTP Tea Party <i>Nancy, France</i>	2024
	Design of a Tableau-Based Automated Theorem Prover and Output of Machine-Checkable Proofs	
	• EuroProofNet School on Natural Formal Mathematics <i>University of Bonn, Bonn, Germany</i>	2025
	• Chair of Theoretical Computer Science Seminar <i>FAU, Erlangen-Nuremberg, Germany</i>	2024
	• VeriDis team seminar <i>Loria, University of Nancy, France</i>	2024
	• LARA team seminar <i>École Polytechnique Fédérale de Lausanne (EPFL), Switerland</i>	2024
	• IGG team seminar <i>ICube, University of Strasbourg, France</i>	2024
	Goéland: a Concurrent Tableaux-Based Theorem Prover	
	• AVM2023 <i>Prague, Czech Republic</i>	2023
	• PDAR2022 <i>Haifa, Israel</i>	2022
	• IJCAR2022 <i>Haifa, Israel</i>	2022
	Formal Method: The Art of Using Logic to Build Safer Systems	
	• Chair of Theoretical Computer Science <i>Faculty of Informatics and Data Science, University of Regensburg, Germany</i>	2023
	Reasoning Methods in Automated Theorem Proving	
	• BOREAL team seminar <i>LIRMM, University of Montpellier, France</i>	2023
	Who Killed Agatha?	
	• PhD seminar <i>LIRMM, University of Montpellier, France</i>	2022
	A Concurrent Tableaux Proof-Search Procedure	
	• MaREL team seminar <i>LIRMM, University of Montpellier, France</i>	2022
	• PhD seminar <i>LIRMM, University of Montpellier, France</i>	2021
	• Proof day <i>LIRMM, University of Montpellier, France</i>	2021

PRIZES AND DISTINCTIONS

- | | |
|---|---|
|  | 3rd Prize - 3MT 2023
<i>French edition of 3 minutes thesis</i> Nîmes, France
Contest in which each candidate must popularise his thesis in 3 minutes. I won the 3 rd prize at the regional final. |
|  | 1st Prize - 5 Minutes to Convince 2023
<i>University of Montpellier</i> Montpellier, France
Contest in which each candidate must present an innovant project in 5 minutes. I won the 1 st price at the PhD category. |
|  | Woody Bledsoe Award 2022
<i>IJCAR2022</i> Haïfa, Israel
Student grant won at IJCAR2022 for the paper "Goéland: a Concurrent Tableau-Based Theorem Prover (System Description)" |
|  | Best Newcomer Prover 2022
<i>CASC2022</i> Haïfa, Israel
Award for the best new prover at CASC, a prover competition. |

SCIENCE PROMOTION

- | | |
|---|--|
|  | Take a Step Back to Code Better 2026
<i>Science's Night (Nuit de la science)</i> Metz, France
Illustration of the concept of reversibility through Legos. |
|  | Introduction to Logic
Introduction to logical reasoning and software verification by solving riddles. <ul style="list-style-type: none"> • <i>National Training Plan</i> University of Lorraine, France 2025 • <i>MATh.en.JEANS</i> University of Lorraine, France 2025 • <i>World Logic Day</i> AoE 2025 • <i>Regional academic delegation to research and innovation</i> LIRMM, France 2022 • <i>LIRMM's open days</i> LIRMM, France 2022 |
|  | Automated Reasoning: Techniques and Applications (a short introduction) 2023
<i>University of Regensburg</i> Regensburg, Germany
Article in a series of books published by the university presenting each of its components. |
|  | The Importance of Popularisation
Promotion of science popularisation through the experience of 3 minute thesis. <ul style="list-style-type: none"> • <i>Science radio programme</i> Divergence FM 2023 • <i>University newsletter</i> University of Montpellier 2023 |
|  | Chiche : 1 scientist, 1 class
Presentation of the researcher's work to high school students. <ul style="list-style-type: none"> • <i>Antoine de Saint-Exupéry high school</i> Fameck, France 2025 • <i>Jules GUESDE high school</i> Montpellier, France 2023 |



Girls and STEM

Exchanges between female high school students and female scientists about computer science, to promote girls in science.

- **Les Cignognes** | University of Lorraine and University of Strasbourg 2025-2026
- **Girls and Maths** | Women and Maths, Animaths 2023
- **MathsC2+** | French Mathematics Society, Ministry of education, Animaths 2022

EVENT ORGANISATION



GT LVP Days 2025-2026
PC Chair | France



GT SCALP Days 2026
Local organizer | Nancy, France



31st International Symposium on Model Checking Software (SPIN) 2025
AE Chair | Hamilton, Canada



Formal Methods in Computer-Aided Design (FM-CAD) 2024
Web Chair | Vienna, Austria



11th Workshop on Horn Clauses for Verification and Synthesis (HCVS24) 2024
PC Chair | Luxembourg City, Luxembourg



1st Summer School of Interactions of Proof Assistants and Mathematics 2023
Volunteer Student | Regensburg, Germany



Session of National Council of Universities, 27th Section (CNU27) 2022
Volunteer Student | Montpellier, France



The 11th International Colloquium on Graph Theory and combinatorics (ICGT) 2022
Volunteer Student | Montpellier, France



The 20th International Conference on Software & Systems Reuse (ICSR) 2022
Volunteer Student | Online



30th Anniversary of LIRMM 2022
Volunteer Student | Montpellier, France



PhD Seminar of LIRMM 2021, 2022
Volunteer Student | Montpellier, France

REVIEWS



PC Member

- EACSL Annual Conference on Computer Science Logic (CSL) 2027
- Certified Programs and Proofs (CPP) 2027
- International Conference on Logic for Programming, Artificial Intelligence and Reasoning (LPAR) 2026
- International Workshop on the Implementation of Logics (IWIL) 2026
- Proof Society 2026
- Congress of Software Science (Poster Session) 2026
- Journées Francophones des Langages Applicatifs (JFLA) 2026
- IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS) 2025
- Formal Methods in Computer-Aided Design (Student Forum) (FMCAD Student Forum) 2025
- International Symposium on Theoretical Aspects of Software Engineering (TASE) 2025



Journals

- Science of Computer Programming 2025, 2026
- Journal of Logical and Algebraic Methods in Programming 2025
- Journal of Applied Logic 2025



Subreviewer

- International Joint Conference on Automated Reasoning (IJCAR) 2022, 2024, 2026
- International Conference on Interactive Theorem Proving (ITP) 2025
- International Conference on Automated Deduction (CADE) 2025
- International Symposium on Formal Methods (FM) 2024
- Formal Methods in Computer-Aided Design (FMCAD) 2024
- International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI) 2024
- Certified Programs and Proofs (CPP) 2023, 2024, 2025



Artifact Reviewer

- International Conference on Computer Aided Verification (CAV) 2026
- International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI) 2025
- International Symposium on Formal Methods (FM) 2024



University of Lorraine

- Functional Programming 2024-2027
Third year of bachelor in computer science
- Agent-Oriented Programming 2025-2027
Second year of bachelor in computer science
- System: processes, memory and files 2024-2027
Second year of bachelor in computer science
- Algorithms and Imperative Programming 2024-2027
First year of bachelor in computer science
- Introduction to Databases 2026-2027
First year of bachelor in medecin
- Software Analysis and Design 2024-2026
Fist year of master's degree in computer science
- Student's projects supervision 2024-2026
Second year of bachelor in computer science
- Logic 2024-2025
Second year of bachelor in computer science



University of Regensburg

- Programming II 2023-2024
First year of bachelor in computer science
- Introduction to Theoretical computer sciences 2023-2024
First year of bachelor in computer science



University of Montpellier

- Program Verification 2022-2023
Third year of bachelor in computer science
- Functional Programming 2022-2023
First year of bachelor in computer science
- Data Warehouse and Big Data 2021-2022
First year of master in computer science
- First-Order Logic 2021-2022
Third year of bachelor in computer science
- Network and Concurrent Programming 2021-2022
Third year of bachelor in computer science
- Parallel and Distributed Programming 2020-2021
First year of master in computer science
- Network, System and Web 2020-2021
First year of bachelor in computer science



PhD Supervision

- Florent KRASNOPOL 2025-2028
Automated Theorem Proving over the Reals for Reasoning on Quantum Circuits | With Stephan MERZ and Sophie TOURRET



Master's Thesis Supervision

- Johann ROSAIN 2021-2022
Indexing Structures for First-Order Logic | With Engel LEFAUCHEUX and Marie DUFLOT-KREMER
- Claude LEROY 2025-2026
Decentralized Diagnosis of Stochastic Systems | With Engel LEFAUCHEUX



Bachelor's Thesis Supervision

- Johann ROSAIN 2021-2022
Deduction Modulo Theory and Polymorphism in Goéland | With Hinde Lilia BOUZIANE, David DELAHAYE and Simon ROBILLARD
- Cédric BERTHET, Enzo GOULESQUE, Lorenzo PUCCIO, 2021-2022
Margaux RENOIR, Tom SIMULA
Arithmetic in Goéland | With Hinde Lilia BOUZIANE, David DELAHAYE and Simon ROBILLARD



Internship Supervision

- Ruka HLUPIC 2026
Automated Reasoning for Nikoli's Games | 2nd year of bachelor in computer science | With Florent KRASNOPOŁ
- Corentin MINAROLI 2026
Unification in Goéland | 3rd year of bachelor in computer science
- Nicolas SCHUTZ-KUCHLY 2026
Equality Reasoning in Goéland | 3rd year of bachelor in computer science
- Titouan LE PEN 2025
Development of an Automated Theorem Prover | 2nd year of bachelor in engineering
- Achille RAZAFIMAHARO 2025
Development of an Automated Theorem Prover | 2nd year of bachelor in engineering
- Filip JAGIELLOWICZ 2024
Implementation of a Decision Procedure for CaAL | 1st year of master in computer science | With Philipp RÜMMER
- Dylan BETTENDROFFER 2023
A Dedukti Output for Goéland | 2nd year of master in computer science | With Hinde Lilia BOUZIANE, David DELAHAYE and Simon ROBILLARD
- Johann ROSAIN 2023
Deskolemization in First-Order Logic | 3rd year of bachelor in computer science | With Hinde Olivier HERMANT and David DELAHAYE
- Matthieu PIERRET 2023
Interactive Proof in Goéland | 2nd year of bachelor in computer science | With Hinde Lilia BOUZIANE, David DELAHAYE and Simon ROBILLARD
- Lorenzo PUCCIO 2022
A Rocq Output for Goéland | 3rd year of bachelor in computer science | With Hinde Lilia BOUZIANE, David DELAHAYE and Simon ROBILLARD
- Adrien MECIBAH 2022
Interactive Traces for ATP | 2nd year of bachelor in computer science | With Hinde Lilia BOUZIANE, David DELAHAYE and Simon ROBILLARD
- Nina JANEVA 2021
Automated Tool for Benchmark | 3rd year of bachelor in computer science | With Hinde Lilia BOUZIANE, David DELAHAYE and Simon ROBILLARD
- Johann ROSAIN 2021
Code Trees for Unification | 2nd year of bachelor in computer science | With Hinde Lilia BOUZIANE, David DELAHAYE and Simon ROBILLARD

COLLECTIVE TASKS



Co-organizer of the ProoVer Competition Since 2026
The TPTP World



Co-head of GT LVP (Languages and Program Verification Working Group) Since June 2025
Software Sciences Research Group (GDR SciLog)



Editor of the AAR newsletter Since June 2024
Association for Automated Reasoning

- 

Mentor Teacher

Faculty of Science and Technology | *Nancy, France*

Individual student monitoring for 1st year bachelor's students in computer science.

2025-2026
- 

Jury CAPES NSI

Ministry of National Education, Higher Education and Research

2025-2026
- 

Speaker at Recruitment Preparation Workshops

French Research Group Programming and Software Engineering (GDR GPL)

2024
- 

Contribution to the Team's Website

Faculty of Informatics and Data Science | *Regensburg, Germany*

Addition of articles and various updates

Sept. 2023 – Aug. 2024
- 

Research Group HRS4R

University of Montpellier | *Montpellier, France*

Reflection group on the needs of researchers in the scope of the "HR Excellence in Research" label.

Dec. 2022 - May 2024
- 

PhD Council of the Laboratory

LIRMM | *Montpellier, France*

Organisation of scientific and cultural activities for the laboratory's doctoral students.

Mar. 2022 - Sept. 2023
- 

Doctoral School Council (I2S, ED166)

I2S | *France*

Doctoral students' representative in the doctoral school council.

Jun. 2021 - Sept. 2023
- 

Laboratory Council

LIRMM | *Montpellier, France*

Doctoral students' representative in the laboratory council.

Oct. 2020 - Sept. 2023

PROFESSIONNAL EXPERIENCES

- 

Project Leader in Clinical Supply Chain

Sanofi | *Montpellier, France*

 - Project leader of the software migration for translation of drug leaflets.
 - Data visualisation and criticality analysis of the application park.
 - Documentation and validation strategies.

Aug. 2019- Aug. 2020